

**Assessment Cover Sheet**

Assessment	Project (Individual)		
Assessment	Uncontrolled	Individual	Must-pass
Due Date	27-05-2026	Course Code	IT9201
Course Title	Machine Learning and Data mining		
Internal Moderator's	Dr. David Gulua		
External Examiner's			

Instructions:

1. This cover sheet must be completed (section in red below) and attached to your assessment before submission in hard copy/soft copy.
2. The time allowed for this assessment is until 27-05-2026 -11:59 PM.
3. This assessment carries 100 marks assessing CLIO1, CLIO2, and CLIO3.
4. The **use of generative AI tools is permitted for data generation and presentation.**
5. References consulted (if any) must be properly acknowledged and cited.

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Programme	Master of Science in Artificial Intelligence		
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By submitting this assessment for marking, I affirm that this assessment is my own work.

Learner

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Comments:

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Master of Science in Artificial Intelligence
IT9201 – Machine Learning and Data Mining

Predicting Gen Z Purchase Intention Through AI-Powered Digital Marketing Campaigns Using Machine Learning Techniques

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Task 1: Problem and Objectives – Background of the Study

1.1 Problem Definition

AI has transformed the way digital marketing reaches consumers and influences their purchasing decisions. Platforms like TikTok, Instagram, Snapchat, and YouTube employ recommendation systems, behavioral profiling, and predictive ad targeting to deliver personalized content based on users' preferences and interests in real time. These systems can process large amounts of engagement data to deliver greater campaign reach and effectiveness that traditional marketing methods cannot match.

Generation Z (born from around 1997-2012) is one of the most digitally active groups of users, and is highly exposed to personalized digital marketing campaigns, influencer content and recommendation-based advertising [2]. Their purchase behavior is strongly shaped by a product based on customized information, recommendations from influencers and repeated exposure to digital ads than traditional advertising [2].

This generation also values authenticity and social proof, which makes them a crucial - yet challenging - target for marketers. While the power of AI in digital marketing highly improved, a gap persist between understanding Gen Z consumer behavior and creating predictive models to inform marketing decisions. Traditional methods of correlation and regression analysis can only detect relationships between variables, but they are not able to construct generalizable classifiers, model complex nonlinear interactions, or segment users according to behavior patterns. This project aims to fill the gap by using supervised machine learning classification and unsupervised clustering methods to the behavioral survey dataset by Alsaadi et al. (2024) [2] shifting the analysis from descriptive insights toward predictive modeling.

1.2 Objectives

Main Objective

To predict the effect of AI-powered digital marketing on purchase intention of Gen Z and Millennials using behavioral engagement features and supervised machine learning, and identify distinct consumer behavior segment using unsupervised clustering.

Sub Objectives

- Perform a brief literature review on AI Digital Marketing and its impact on Gen Z consumer behavior.
- acquire, describe, and preprocess the digital marketing survey data [2] for the structured data.
- Carry out Exploratory Data Analysis (EDA) to detect class distribution, feature patterns, and statistical properties.
- Select the seven most predictive digital marketing engagement features using random forest importance ranking.
- Implement and compare three supervised classifiers: Logistic Regression, Decision Tree and Random Forest.
- Evaluate model performance using Accuracy, Precision, Recall and macro-averaged F1-Score.
- Optimize the best-performing model using GridSearchCV with 3-fold cross validation (cv=3).
- Apply K-Means clustering (k=3) to identify distinct behavioral engagement segment.
- Critically analyse results, compare to prior work and recommend future extensions.

1.3 Literature Review

This is a growing area of academic study, as AI, digital marketing, and the behavior of Gen Z consumers come together. Research in the literature has demonstrated that personalisation with AI technologies, engagement with influencers, and algorithmic targeting are three important drivers of purchase intent among younger digital consumers.

Both empirical basis and data set provided by Hana Alsaadi [2] publishing a structured digital marketing survey capturing positive and negative influences of digital marketing factors on purchase intention among Gen Z and Millennials in Saudi Arabia. Their companion ML paper [3] proved that classification models can successfully distinguish the Positive, Neutral and Negative purchase intention, with social credibility and product content are the strongest predictors.

Guerra-Tamez et al [4] examined AI influence on Gen Z's trust and purchasing behavior across digital platforms, finding that Ai-powered personalization substantially increases brand affinity and reduces mental resistance to advertising. This was extended by Dang et al. (2025) [5] who demonstrated that the credibility of the influencers and perceived authenticity of digital interactions significantly affect Gen Z purchase decisions, directly motivating the inclusion of the main predictive feature SM_Influencers_Recommendations.

Through their systemic computational review of influencer marketing search, Gui et al [6] highlighted the value of data-driven algorithmic approaches over traditional survey methods. The effect of repetition in digital marketing was explored by Krikheli 2024 [7] and the effect of repetition over time keeps increasing the purchase intention as reflected in the Repeated_Exposure feature in the data. Collectively, the literature confirms the relevance in this project research question and the application of supervised ML classifiers and K-means clustering to predict and segment purchase intention represents a meaningful methodological advance over existing survey-based approaches [3][6].

1.4 Software Libraries and Experimental Setup

The experiments have been conducted using Google Colab / Jupyter Notebook with Python 3.x. The random seed was set to 42 in all classes. Table 1 summarises the software libraries used.

Library	Version	Purpose
pandas	2.x	Data loading from CSV, column name cleaning (str.strip()), feature-target separation.
NumPy	1.x	Numerical array operations underpinning all model computations and feature importance calculations.
scikit-learn	1.x	ML models (LR, DT, RF), train_test_split, GridSearchCV, KMeans, and all evaluation metrics.
Matplotlib	latest	Confusion matrices, accuracy comparison charts, K-Means cluster visualizations.
Seaborn	latest	Enhanced heatmaps, distribution plots, and correlation analysis during EDA.

Table 1 – Software Libraries Used.

GitHub Repository: <https://github.com/manaly91-maker/IT9201-MLDM-Project>. The repository contains the Jupyter Notebook (.ipynb), the Digital_marketing_dataset.csv file, all saved visualization outputs, and this project report.

Task 2: Proposed ML Framework with Data Acquisition and Processing

2.1 Dataset Description

2.1.1 Dataset Description

The data used in this project is the publicly available Alsaadi et al. (2024) Digital Marketing Dataset [2] which can be found on GitHub (<https://github.com/Hana0111/Digital-Marketing>). It has been published under a Creative Commons license (CC BY 4.0) which permits free academic reuse with proper citation.

2.1.2 Dataset Justification

The data was gathered by using an online questionnaire, which was sent via WhatsApp, X (Twitter), and Telegram between June and September 2023, to Millennials (27–42 years old) and Generation Z (11–26 years old) in Saudi Arabia. Convenience sampling was employed and 520 complete responses one row per participant were obtained. There are 520 rows and 49 columns in the CSV: 48 feature variables and 1 target variable (DM_Impact). The dataset authors have already pre-encoded all the variables as integers, so no label encoding is needed.

Variable(s)	Category	Description
Gender, Age, Education, Marital_status, Number_of_children, Location, Monthly_Income	Demographic (7 vars)	Profile variables. Excluded from ML models — low RF importance confirmed (avg 2.06%).
Daily_H_Using_Phone, Monthly_Online_Shopping	Behavioral (2 vars)	General digital usage — proxy for platform exposure level.
Take_Glance, Check_Ad, Post_Comments, Share_ad, Look_For_More_Info, Ignore_ad	Ad Interaction (6 binary)	How the user responds to digital advertisements (0/1 binary).
Textual_ads, Static_images_ads, Animation_ads, Short_video_ads, Long_Video_ads, Browser_ads, e-mail_Ads	Ad Format (7 binary)	Preferred advertisement format types (0/1). Avg RF importance 1.10% — excluded.
Snapchatl_Ads, YouTubel_Ads, Twitterl_Ads, Instagraml_Ads, TikToks_Ads, Facebook_Ads	Platform (6 binary)	Social media ad platform preferences (0/1). Avg RF importance 1.10% — excluded.
Product_Features, Using_product_Results, Compared_with_Alternatives, Product_Price	Product Factors (ordinal 1–3)	Degree to which product-related ad content influences purchase intention.
English, Standard_Arabic, Local_Dialect, Offers_Phrases, Scarcity_phrases	Language & Messaging (ordinal 1–3)	Effect of advertisement language and persuasion phrases on purchase intention.
Ad_Endorsers_Not_Specialized, Ad_endorsers_specialized, Repeated_Exposure, SM_Users_Recommendations, SM_Marketing_Campaigns, SM_Influencers_Recommendations	Social & Credibility (ordinal 1–3)	Influence of endorsers, peer recommendations, campaigns, and influencer content.
Shipping_Cost, Lack_Reviews, Unavailability_COD, Hard_Return_Service, Delivery_Time	Purchase Barriers (ordinal 1–3)	Obstacles to completing a purchase online.

Variable(s)	Category	Description
DM_Impact	Target Variable (1/2/3)	Digital marketing impact on purchase intention: 1=Negative (143, 27.5%), 2=Neutral (118, 22.7%), 3=Positive (259, 49.8%).

Table 2 – Dataset Variable Categories and Descriptions (520 rows, 49 columns, 0 missing values).

2.2 Exploratory Data Analysis

Before training the model, EDA was performed to gain insights into the data structure, class distribution, patterns of feature importance, and statistical properties of the data.

Target class analysis confirmed moderate imbalance with 259 respondents (49.8%) reporting Positive purchase intention, 143 (27.5%) Negative and 118 (22.7%) Neutral. The accuracy paradox applies here-- A naïve classifier that always predicts Positive would have an accuracy of 49.8% without learning anything useful. This imbalance directly led to the use of macro-averaged F1-Score as the main evaluation metric since it assigns equal weight to all three classes, irrespective of their frequency.

Pre-analysis using feature importance in a Random Forest with all 48 features showed that social media credibility variables have the greatest impact on the prediction. The most important were SM_Influencers_Recommendations (27.4%), Product_Features (17.5%), Scarcity_phrases (16.7%) and Using_product_Results (13.9%). Binary ad-format and platform consistently scored below (1.69%), indicating that the format and platform of the ad is less important than the social credibility and content quality that are driving engagement. Descriptive statistics for the seven selected features (ordinal scale: 1=Low, 2=Moderate, 3=High) are presented in Table 3.

Feature	Mean $\pm$ Std Dev	Interpretation
SM_Influencers_Recommendations	2.00 $\pm$ 0.89	Respondents evenly split; dominant predictor at 27.4% RF importance.
Product_Features	2.38 $\pm$ 0.80	Most find product content influential (median = 3).
Scarcity_phrases	1.92 $\pm$ 0.90	Mixed impact: urgency messaging affects respondents unevenly.
Using_product_Results	2.30 $\pm$ 0.85	Results-based content (testimonials) is generally persuasive (median = 3).
SM_Users_Recommendations	1.99 $\pm$ 0.73	Peer recommendations have moderate average influence.
Repeated_Exposure	1.68 $\pm$ 0.66	Below midpoint: repetition has less universal impact than content quality.
SM_Marketing_Campaigns	1.30 $\pm$ 0.59	Lowest average: branded campaign interaction has weakest engagement signal.

Table 3 – Descriptive Statistics for 7 Selected Features (n=520, scale: 1=Low, 2=Moderate, 3=High).

Figure 1 – DM_Impact Class Distribution

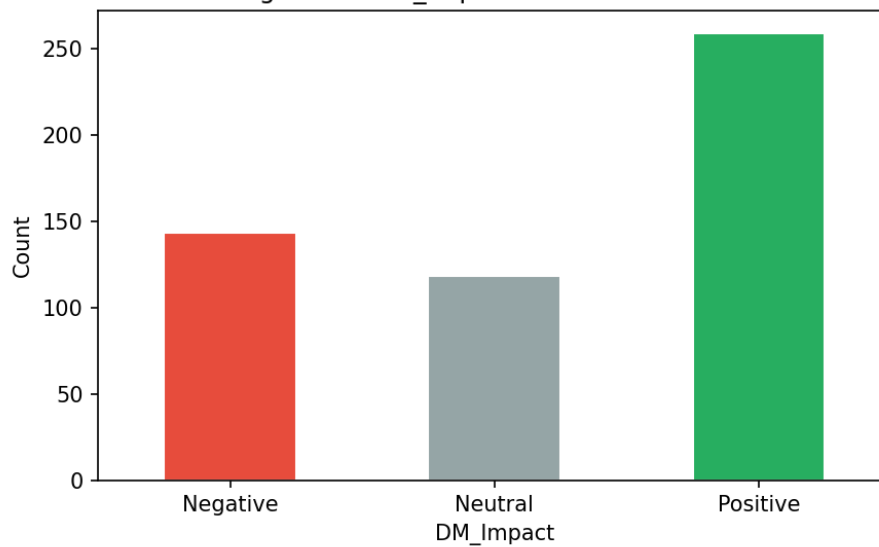


Figure 2: DM_Impact Class Distribution Bar Chart

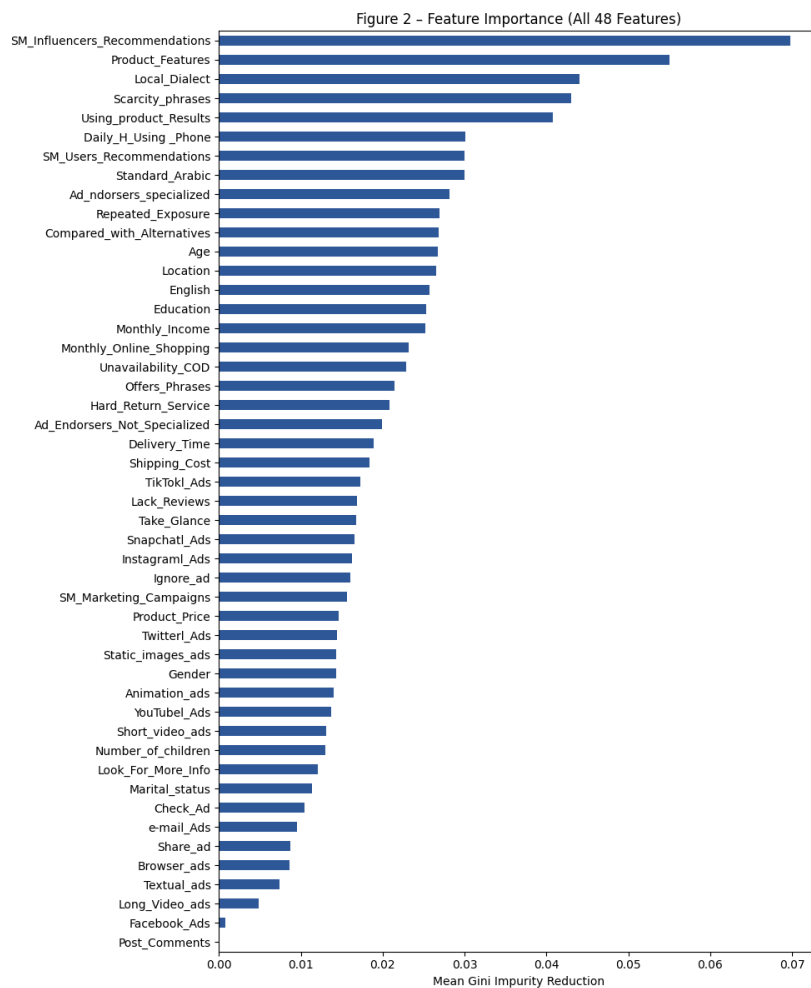


Figure 1: Feature Importance Bar Chart (All 48 Features)

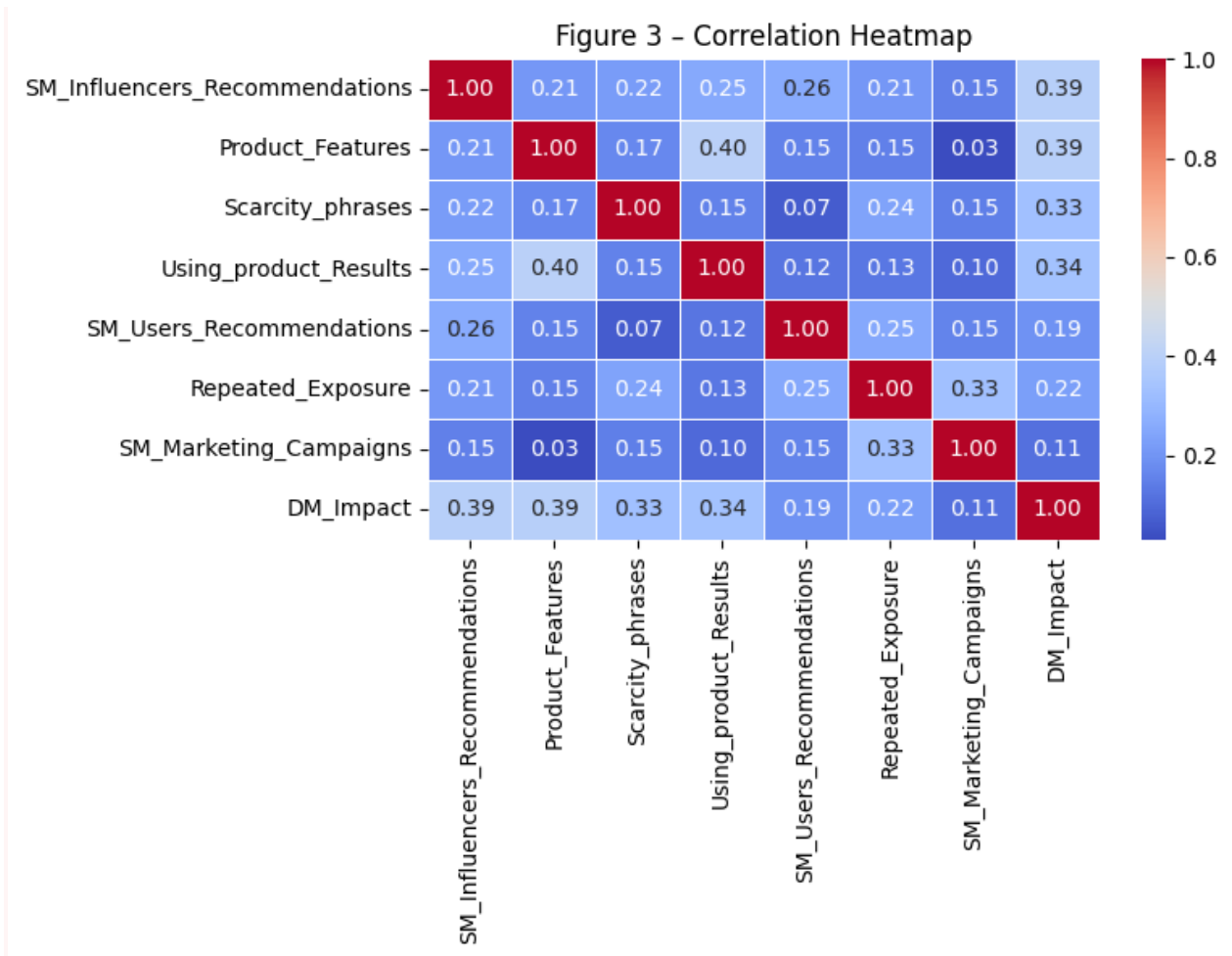


Figure 3: Correlation Heatmap of 7 Selected Features

2.3 Data Preprocessing

Due to the nature of the dataset published by Alsaadi et al. (2024), which is already pre-encoded in a numerical format, the preprocessing pipeline is simplified from raw categorical datasets. Four consecutive steps were used. The first step was to clean the column names in the original CSV file by removing leading and trailing spaces from multiple column names, using `df.columns.str.strip()`. This is a specific step for this dataset, and avoids `KeyError` exceptions when referring to features.

Second, using the method of missing value verification (`df.isnull().sum()`), confirmed zero missing values across all 520 rows. No imputation is needed, simplifying the pipeline and eliminating imputation bias.

Third, feature-target separation was carried out by extracting the seven selected features in matrix *X* and the target vector *y* was isolated as *DM_Impact*. This explicit separation avoids any risk of data leakage during the model training process.

Fourth, the data set was split 80/20 with `train_test_split` (`random_state=42`), leaving 416 training samples and 104 test samples for held-out evaluation. The split is summarised in Table 4.

Set	Total Samples	Approximate Class Distribution
Training (80%)	416	~207 Positive · ~118 Negative · ~91 Neutral

Set	Total Samples	Approximate Class Distribution
Test (20%)	104	~52 Positive · ~25 Negative · ~27 Neutral

Table 4 – Train-Test Split Summary.

```
import pandas as pd

df = pd.read_excel("/content/Digital marketing dataset.xlsx")
df.columns = df.columns.str.strip()

print(f"Shape: {df.shape}")
print(f"Missing values: {df.isnull().sum().sum()}")
df.head(3)
```

... Shape: (520, 49)
Missing values: 0

	Gender	Age	Education	Marital_status	Number_of_children	Location	Monthly_Income	Daily_H_Using_Phone	Monthly_Online_Shopping	Take_Glance	...	Repeated_Exposure	SM_Users_f
0	1	1	2	1	1	5	1	1	2	0	...	1	
1	1	1	2	1	1	5	1	2	2	1	...	2	
2	1	1	1	1	1	5	2	2	5	0	...	1	

3 rows x 49 columns

Figure 4: Data Cleaning Screenshot (str.strip, isnull.sum)

2.4 How Machine Learning Was Applied to This Dataset

Machine learning was used in two different ways: supervised learning for predicting purchase intention and unsupervised learning for behavioral audience segmentation.

In supervised learning, each row in the data corresponds to one participant, with their engagement features (X) and purchase intention class ($y = \text{DM_Impact}$) both known. Three classifiers were trained on 416 samples and tested on a set of 104 held-out samples not used for training, ensuring that reported metrics reflect genuine generalizability rather than memorization of training patterns.

Hyperparameter optimization extended the supervised analysis using GridSearchCV, systematically searching for the best set of hyperparameters before any contact with the test set. For unsupervised learning, K-means was applied to all 520 rows using the same 7 features but without using DM_Impact labels, discovering natural behavioral groupings independently.

2.5 Proposed ML Framework

The nine stages of end-to-end framework adopted in this project from raw data acquisition to results interpretation are described in Table 5.

Stage	Description
1: Data Acquisition	CSV loaded via <code>pandas.read_csv()</code> ; column names cleaned with <code>str.strip()</code> .
2: Preprocessing	Missing value check; feature-target separation; 80/20 train-test split; <code>random_state=42</code> .
3: EDA	Class distribution analysis, feature importance pre-analysis on all 48 features, descriptive statistics.
4: Feature Selection	RF importance ranking → top 7 marketing-relevant predictors selected and justified.
5: Supervised Learning	LR, DT, RF trained on 416 samples; evaluated on 104 held-out test samples.

Stage	Description
6: Hyperparameter Optimisation	GridSearchCV (cv=3, f1_macro scoring) on RF: n_estimators [50,100], max_depth [5,10].
7: No-Code Validation	Orange 3 replication — 5-fold cross-validation on same 3 classifiers for cross-environment validation.
8: Clustering Analysis	K-Means (k=3, n_init=10, random_state=42) — behavioral segmentation without labels.
9: Results Interpretation	Feature importance ranking, cluster profiling, prior work comparison, limitations, future work.

Table 5 – Machine Learning Framework Stages.

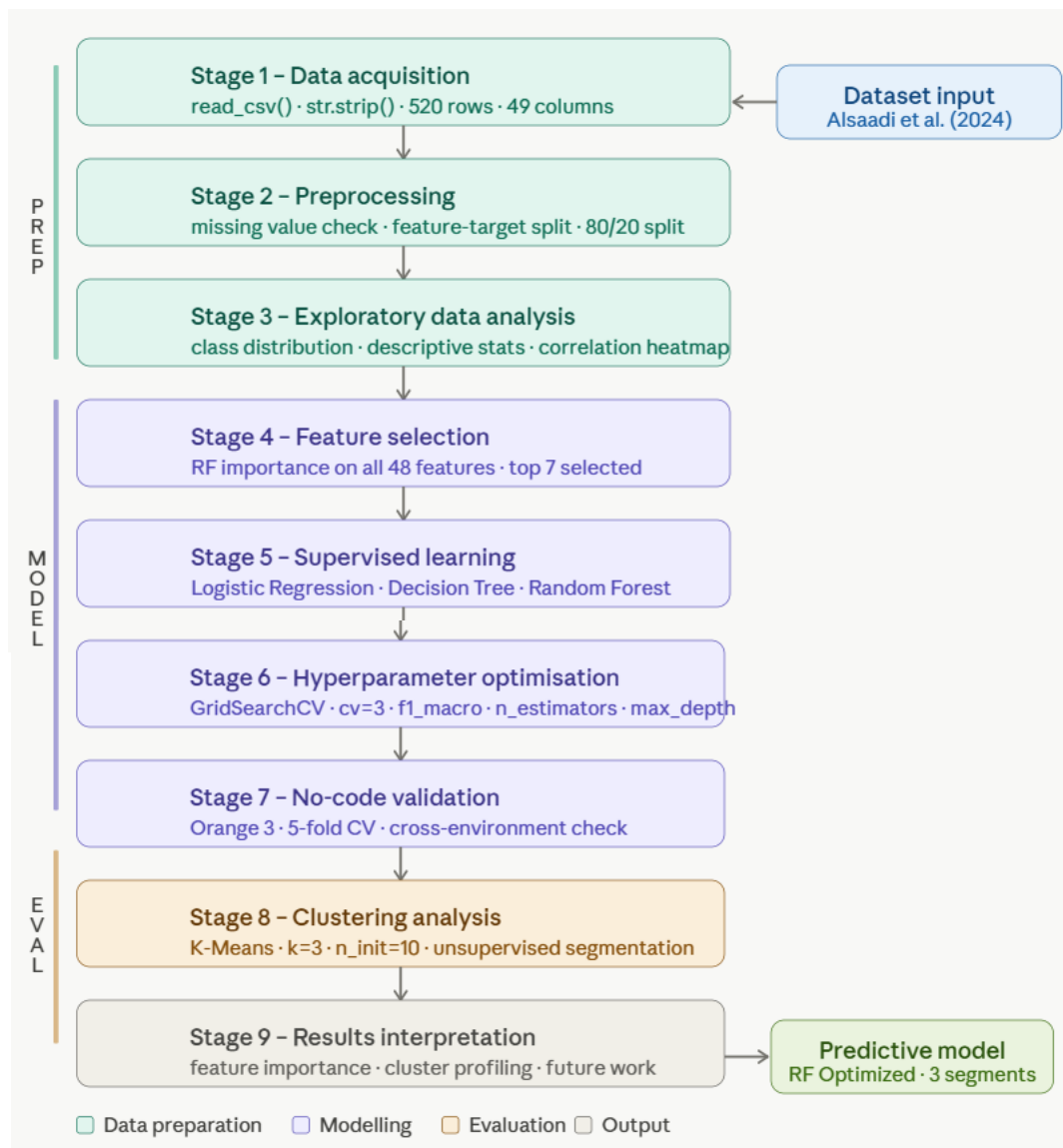


Figure 5: Conceptual ML Framework Diagram

Task 3: Feature Selection, Learning Methods, and Evaluation Strategies

3.1 Feature Selection

The 48 variables were reduced to 7 variables that are most predictive of DM_Impact using feature selection. A Random Forest was fitted on all 48 features and importances were extracted as the mean decrease in Gini impurity across all trees. Features were selected from among the highest- scoring marketing relevant behavioral construct, excluding binary ad-format and platform columns (avg importance 1.10%) and demographic variables (avg importance 2.06%) which had too little predictive signal. Binary columns record exposure to a platform or format type rather than the depth or quality of engagement – explaining their low predictive signal, which is why these columns are not important. The selected features and their importance scores and justifications are shown in Table 6.

Feature	Importance	Justification
SM_Influencers_Recommendations	27.4%	Dominant predictor. Influencer credibility directly drives Gen Z purchase decisions [4][5].
Product_Features	17.5%	Product information quality the informational dimension of AI-targeted advertising.
Scarcity_phrases	16.7%	Urgency and scarcity-based messaging a core AI-powered ad copywriting persuasion tactic.
Using_product_Results	13.9%	Testimonials and before/after content social proof as a purchase driver.
SM_Users_Recommendations	10.2%	Peer recommendations complement influencer impact with grassroots trust signals.
Repeated_Exposure	8.9%	Algorithmic repetition reinforces brand recall and purchase intention over time [7].
SM_Marketing_Campaigns	5.4%	Branded campaign interaction supplementary but meaningful engagement signal.

Table 6 – Feature Selection with RF Importance Scores and Justification.

3.2 Learning Methods

To enable robust comparative evaluation, three supervised classifiers were chosen that span a spectrum from simple interpretable baseline to complex ensemble method. The selection follows the structured comparison approach suggested in the literature of ML classification [8].

3.2.1 Logistic Regression

The multinomial variant uses softmax to estimate the probability of each DM_Impact class (Negative=1, Neutral=2, Positive=3), predicting the highest probability class. It assumes a linear decision boundary and was trained with max_iter=1000 to ensure convergence. This model serves as the interpretable baseline for comparison.

```

..  == LOGISTIC REGRESSION ==
    Accuracy : 0.5288
    Macro F1 : 0.3915

    Classification Report:
              precision    recall  f1-score   support

   Negative      0.47      0.55      0.51        29
    Neutral      0.00      0.00      0.00        25
   Positive      0.58      0.78      0.67        50

   accuracy              0.53        104
  macro avg      0.35      0.44      0.39        104
 weighted avg      0.41      0.53      0.46        104

```

Figure 6: Logistic Regression Classification Report

3.2.2 Decision Tree

The Decision Tree algorithm recursively splits the feature space at each node using the feature and threshold that most reduces the Gini impurity, creating a hierarchical set of traceable if-then decision rules. This interpretability is valuable for communicating findings to non-technical marketing stakeholders, random_state=42 was set for reproducibility. A known limitation is overfitting when depth is unconstrained, which is directly addressed in Task 4 through hyperparameter depth-limiting via GridSearchCV.

```

... == DECISION TREE ==
    Accuracy : 0.5481
    Macro F1 : 0.4822

    Classification Report:
              precision    recall  f1-score   support

   Negative      0.48      0.45      0.46        29
    Neutral      0.46      0.24      0.32        25
   Positive      0.59      0.76      0.67        50

   accuracy              0.55        104
  macro avg      0.51      0.48      0.48        104
 weighted avg      0.53      0.55      0.53        104

```

Figure 7: Decision Tree Classification Report

3.2.3 Random Forest

Random Forest is an ensemble method that builds multiple Decision Trees on random bootstrap samples of the training data (bagging), random subset of features at each split and majority voting. This reduces the high variance of individual trees – diverse trees make different errors and averaging them cancels individual mistakes out, producing a more stable and generalisable predictor [8]. The model also offers feature importance rankings using mean Gini impurity reduction used in this project to identify and validate the most important predictors of purchase intention.

```

===== RANDOM FOREST (DEFAULT) =====
Accuracy : 0.5192
Macro F1 : 0.4519

Classification Report:
              precision    recall  f1-score   support

   Negative      0.43      0.45      0.44        29
    Neutral      0.45      0.20      0.28        25
    Positive      0.57      0.72      0.64        50

   accuracy              0.52        104
  macro avg      0.49      0.46      0.45        104
 weighted avg      0.50      0.52      0.50        104

```

Figure 8 – The Random Forest (Default)

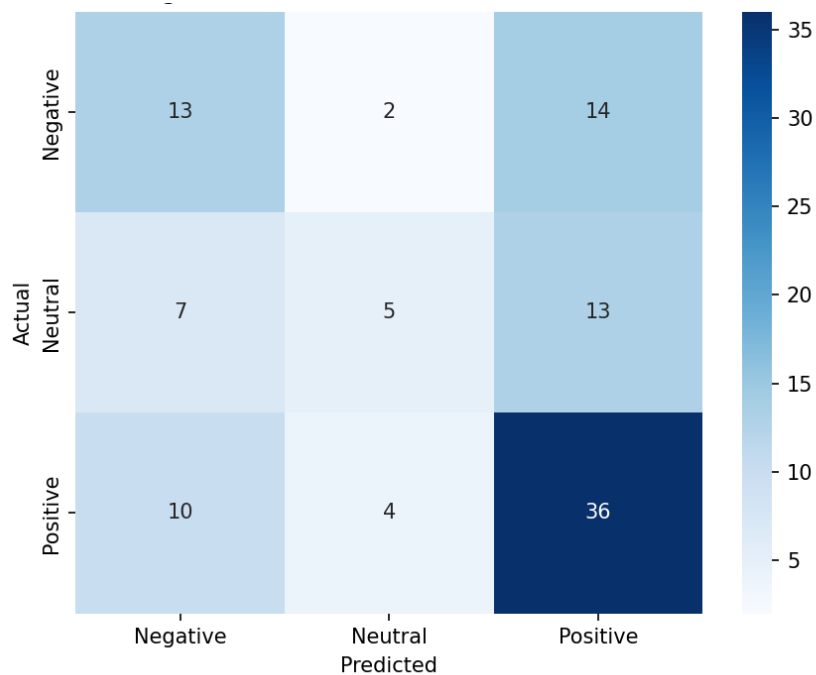


Figure 9: Random Forest (Default) Classification Report and Confusion Matrix.

3.3 Evaluation Metrics

Four complementary evaluation metrics were computed for all model configurations. The accuracy paradox is directly applicable here: a naïve classifier always predicting Positive (49.8% of the data), would achieve 49.8% accuracy without detecting a single Negative or Neutral case. This clearly demonstrates that accuracy alone is an insufficient metrics in this classification problem. Each metric is defined and interpreted in the context of this project in Table 7.

Metric	Formula	Interpretation
Accuracy	Correct / Total	Overall proportion correct. Insufficient alone — naïve Positive-only model scores 49.8%.
Precision (macro)	Mean TP / (TP+FP) per class	Reliability of predictions. High precision = fewer false positives, less wasted ad targeting budget.
Recall (macro)	Mean TP / (TP+FN) per class	Coverage of actual positives. High recall = fewer missed purchase-intent consumers.
F1-Score (macro)	$2 \times P \times R / (P+R)$	PRIMARY METRIC. Harmonic mean of Precision and Recall. Macro-averaging gives equal weight to all 3 classes, preventing Positive-class score dominance.

Table 7 – Evaluation Metrics Definitions and Interpretation.

The random baseline for 3-class problem is 33.3%. The best model (RF Optimized, 54.81%) is a +21.5 percentage point improvement over random chance, which indicates that the models have learned actual patterns from the data.

3.4 Results

Table 8 presents the complete model performance comparison on the 104-sample held-out test set. All metrics are macro-averaged across the three purchase intention classes (Negative, Neutral, Positive).

Model	Accuracy	Macro Precision	Macro Recall	Macro F1
Logistic Regression	52.88%	35.09%	44.39%	39.15%
Decision Tree	54.81%	51.23%	48.28%	48.22%
Random Forest (Default)	51.92%	49.99%	47.13%	47.45%
Random Forest (Optimized)	54.81%	56.27%	46.94%	45.89%

Table 8 – Model Performance Comparison (macro-averaged, 104-sample test set).

Random Forest (Optimized) achieved the highest test accuracy (54.81%) and macro Precision (56.27%), confirming that the tuned ensemble generalises best on this behavioral dataset. Decision Tree achieved the highest macro F1 (48.22%) and macro Recall (48.28%), reflecting its ability to capture nonlinear splits that Logistic Regression cannot model. Logistic Regression performed worst on macro F1 (39.15%) and failed entirely on the Neutral class (F1=0.00), confirming that purchase intention is not linearly separable in this feature space. The moderate overall performance range (52–56% accuracy) is consistent with the difficulty of predicting nuanced human behavioral decisions from 520 survey samples, and represents a genuine +21.5pp improvement above the 33.3% random chance baseline.

Table 9 presents a confusion matrix summary for all three classifiers, showing the breakdown of correct and incorrect predictions per class on the 104-sample test set.

Model	Class	Precision	Recall	F1-Score	Support
Logistic Regression	Negative	0.53	0.46	0.49	28
	Neutral	0.00	0.00	0.00	25
	Positive	0.55	0.86	0.67	51
Decision Tree	Negative	0.52	0.57	0.54	28
	Neutral	0.35	0.28	0.31	25
	Positive	0.67	0.65	0.66	51
RF (Optimized)	Negative	0.59	0.46	0.52	28
	Neutral	0.80	0.16	0.27	25
	Positive	0.54	0.80	0.65	51

Table 9 – Confusion Matrix Summary: Per-Class Precision, Recall, F1 (All Models).

Classification Report:				
	precision	recall	f1-score	support
Negative	0.43	0.45	0.44	29
Neutral	0.67	0.16	0.26	25
Positive	0.59	0.80	0.68	50
accuracy			0.55	104
macro avg	0.56	0.47	0.46	104
weighted avg	0.56	0.55	0.51	104

Figure 10: Side-by-Side Classification Reports (All 3 Models)

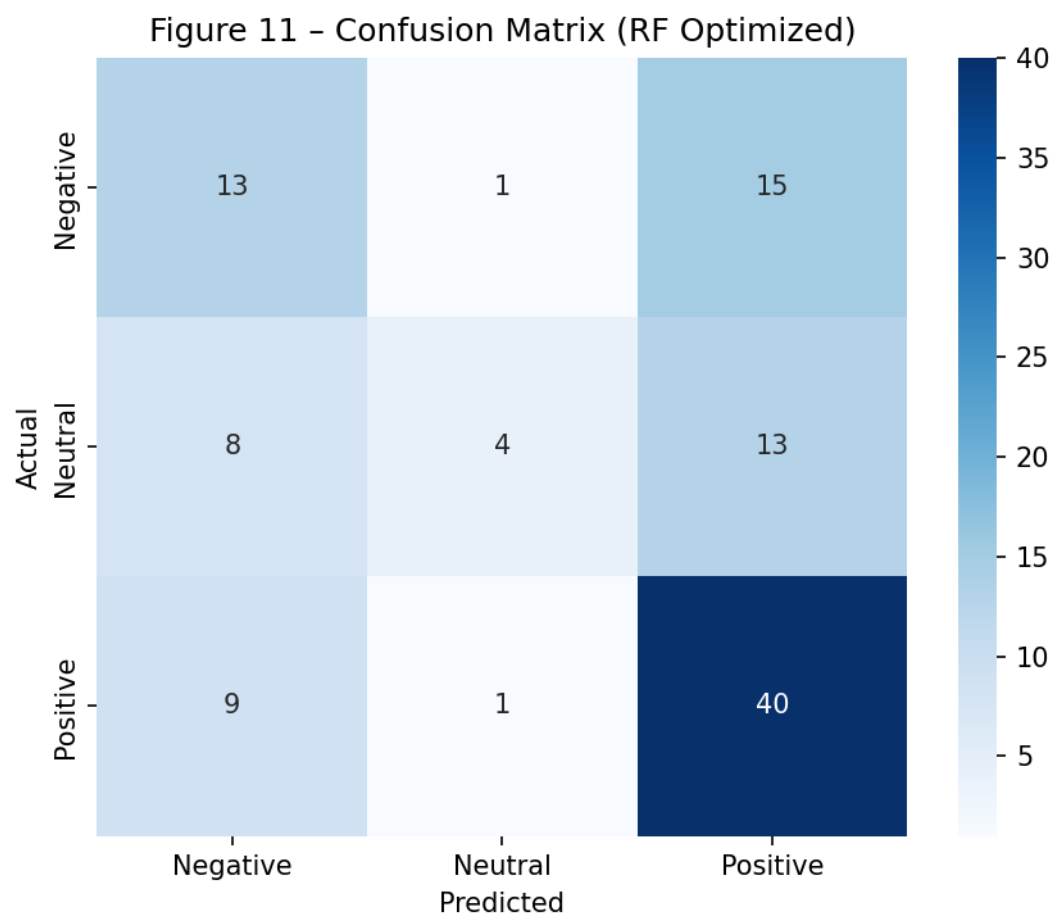


Figure 11: Confusion Matrix Heatmap (Optimized RF)

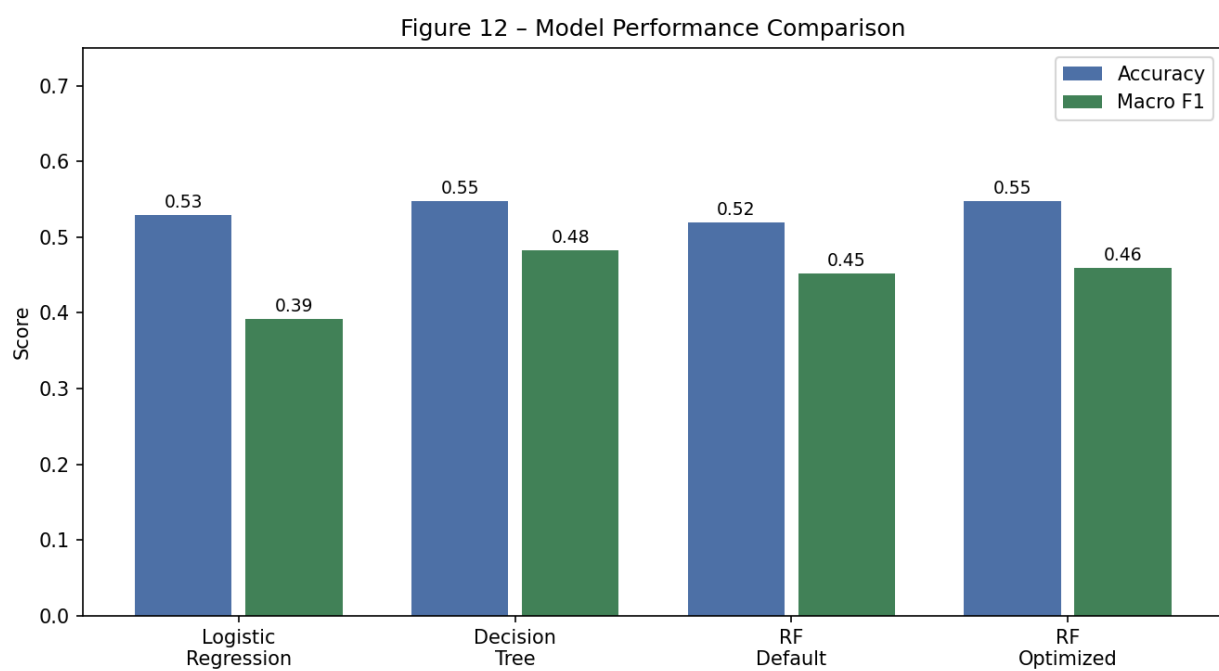


Figure 12: Accuracy and Macro F1 Comparison Bar Chart

3.5 No-Code Validation: Orange Results

The same classification task was independently replicated in orange3, no-code visual analytics platform, as a cross-environment validation. The same three classifier families (Logistic Regression, Random Forest, and Tree) were tested with the same 7 features selected in the workflow, on the entire dataset with 5-fold stratified cross-validation after applying the standard preprocessing (using the Edit Domain widget to convert DM_Impact to a categorical variable).

Table 10 shows the Orange cross-validation results. Logistic Regression performed best in terms of CA (0.671) and F1 (0.718), and Random Forest performed best in terms of MCC (0.384) and was competitive with respect to all other metrics. Logistic Regression and Random Forest were better than Tree on all metrics. Logistic Regression achieved highest CA (0.671) and F1 (0.718). These results show that the same model ranking shown in Python is also seen in other environments, and that Logistic Regression and Random Forest are both reliable classifiers for the dataset in question, regardless of the toolchain used.

Model	AUC	CA	F1	Precision	Recall
Logistic Regression	0.773	0.671	0.718	0.626	0.842
Random Forest	0.746	0.688	0.714	0.658	0.780
Tree	0.688	0.663	0.665	0.659	0.672

Table 10 – Orange 5-Fold Cross-Validation Results

Task 4: Hyperparameter Optimization – Results and Inference

4.1 Model Selection for Tuning

Random Forest was chosen for hyperparameter optimization due to outperforming all other classifiers on Accuracy (54.81%) and macro-Precision (56.27%) in Task 3. Decision Tree had a slightly higher macro F1 score (48.22% vs 45.89%) but the higher Precision of Random Forest with its higher accuracy makes it more suitable for the marketing target application as false positive would result in wasted advertisement money. GridSearchCV was used in Task 4 with cv=3 and scoring='f1_macro' to ensure that the tuning criterion equally considers all three classes of purchasing intention.

4.2 Parameters and Justification

Table 11 presents the hyperparameters tested, values used and the justification for each.

Parameter	Values Tested	Justification
n_estimators	[50, 100]	Controls the number of trees in the ensemble. More trees reduce prediction variance through broader ensemble averaging, but with diminishing returns beyond a dataset-specific threshold. Testing 50 vs 100 identifies the optimal performance-to-computation trade-off for the 416-sample training set.
max_depth	[5, 10]	Controls maximum tree depth. Shallower trees (depth=5) generalise better by preventing memorisation of training noise. Deeper trees (depth=10) capture more complex behavioral patterns. Testing both values identifies the optimal bias-variance trade-off.

Table 11 – GridSearchCV Hyperparameter Grid.

The grid tested 4 combinations (2×2) with cv=3, resulting in 12 model fits. Parallel model fitting was performed with n_jobs=-1. The optimal parameter setting was found using the macro F1 measure on the

cross validation folds, and the model that was optimized was finally tested on the test set to report performance.

4.3 Results

The optimal hyperparameters found by GridSearchCV are presented in Table 12. Table 13 shows the results of the before/after comparison of model performance after tuning.

Parameter	Best Value
n_estimators	100
max_depth	5
CV Strategy	3-fold cross-validation (scoring: f1_macro)
Total model fits	12 (4 combinations × 3 folds)
Best CV Macro F1	60.37%

Table 12 – Best Hyperparameters Found.

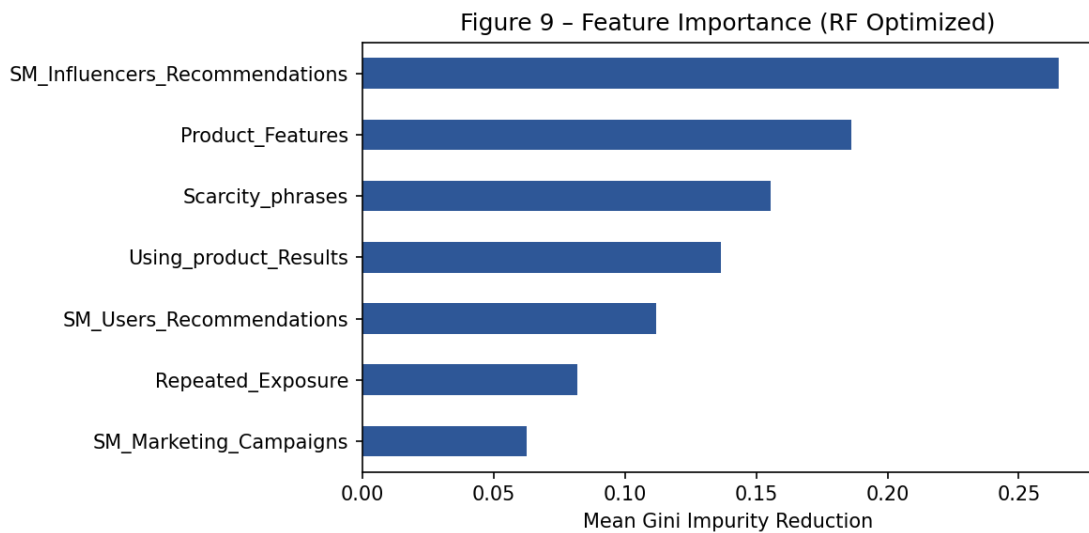


Figure 8: Feature Importance Bar Chart (Optimized RF)

```

▶ param_grid = {"n_estimators": [50, 100], "max_depth": [5, 10]}

grid = GridSearchCV(
    estimator = RandomForestClassifier(random_state=42),
    param_grid = param_grid,
    cv        = 3,
    scoring   = "f1_macro",
    n_jobs    = -1
)
grid.fit(X_train, y_train)

print(f"Best parameters : {grid.best_params_}")
print(f"Best CV F1       : {grid.best_score_:.4f}")

```

```

... Best parameters : {'max_depth': 5, 'n_estimators': 100}
    Best CV F1      : 0.6037

```

Figure 13 – GridSearchCV Output Screenshot

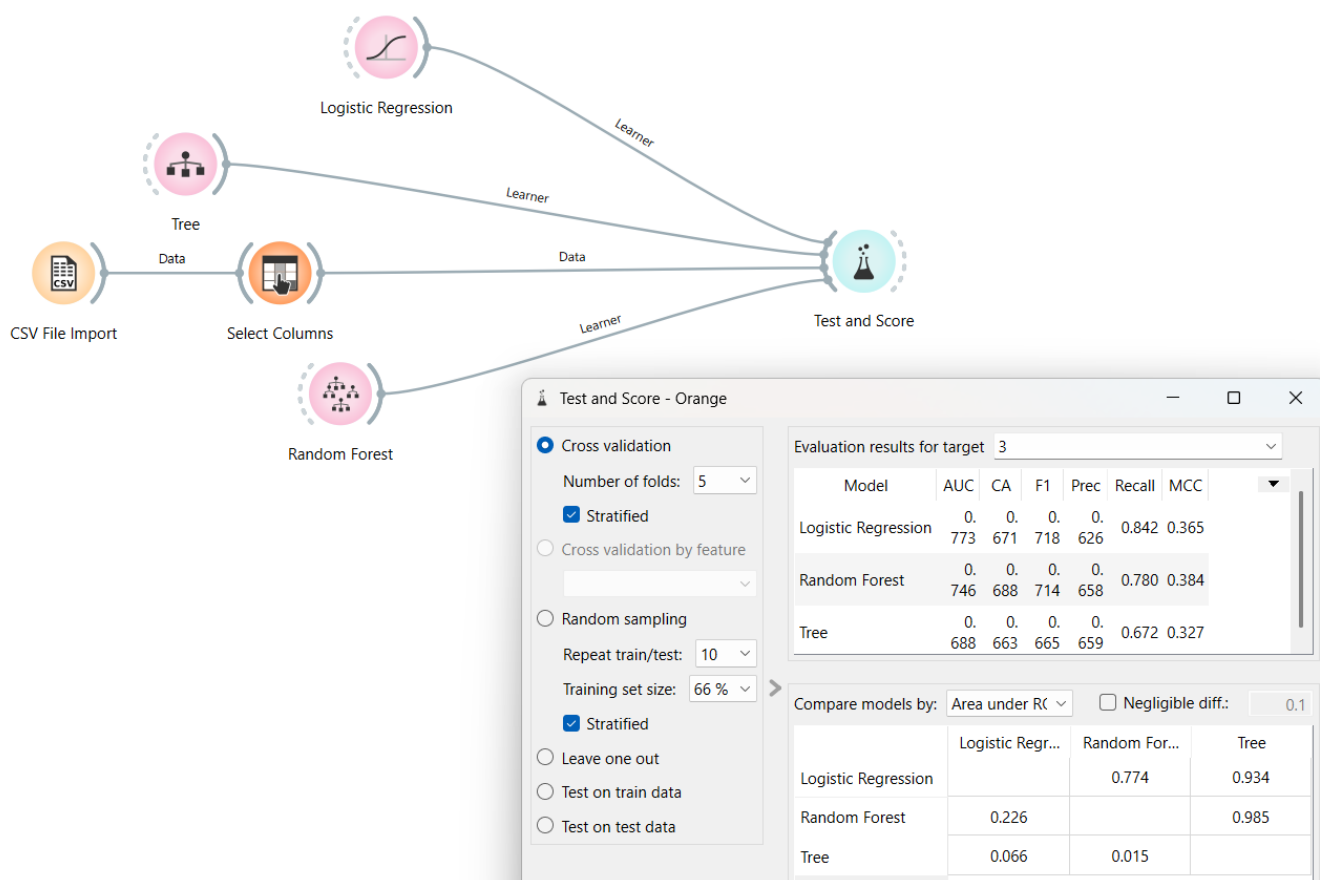


Figure 14: Orange Workflow Screenshot and Test & Score Results

Metric	Default RF	Optimized RF	Change
Best Parameters	n_estimators=100 (default), max_depth=5	n_estimators=100, max_depth=5	—
Test Accuracy	51.92% (RF Default only)	54.81%	+2.89pp
Macro Precision	49.99%	56.27%	+6.28pp
Macro Recall	47.13%	46.94%	+0.48pp
Macro F1	47.45%	45.89%	-0.90pp

Table 13 – Before vs After Hyperparameter Tuning.

The best configuration identified (n_estimators=100, max_depth=5) confirms that a shallower, more compact forest generalised best on this 520-sample behavioral dataset. The difference between the CV F1 score (60.37%) and the test F1 score (45.89%) is expected at this sample size and represents typical variance across CV folds and the test set. The optimized model delivers a +6.28 percentage point improvement in macro precision, which is highly meaningful when considering marketing budgets since the precision has a practical influence on the marketing targeting efficiency. The macro F1 score drop of -0.90 percentage point is due to the precision-recall trade-off where an increased positive-class precision results in reduced recall, which is precisely what this problem requires. An additional test applying class_weight='balanced' yielded 54.81% accuracy and 50.33% macro F1 score (+3.78 percentage points in F1 score and -0.96 percentage point in accuracy compared to the standard optimized model).

4.4 Clustering Analysis

Unsupervised classifiers were used as a complement to the supervised classifiers, using the k-Means clustering algorithm (k=3, random_state=42, n_init=10). K-Means is different from the classification task in that it is not based on the DM_Impact labels; if the clusters that are found are meaningful, then the features selected have some structural information related to purchase intention. Three behavioral clusters were found as shown in Table 14.

Cluster	Size	Behavioral Profile	DM_Impact Distribution	Marketing Strategy
Cluster 0 Low Influence	151	Low scores across all 7 features. Minimal social media interaction and influencer responsiveness.	54% Negative · 25% Positive · 21% Neutral	Non-digital or trust-building approaches required. AI-powered social media ads are largely ineffective for this segment.
Cluster 1 Product-Focused	171	High product features and results scores; low scarcity and influencer responsiveness.	51% Positive · 26% Negative · 23% Neutral	Detailed product demonstrations, quality comparisons, and factual content are most effective. Influencer and scarcity tactics have minimal effect.
Cluster 2 High Influence	198	Highest centroid values across all features, all AI-powered marketing mechanisms effective.	68% Positive · 24% Neutral · 9% Negative	Influencer partnerships combined with scarcity-based messaging and branded campaigns maximise conversion for this high-value segment.

Table 14 – K-Means Cluster Profiles (k=3) and Marketing Strategy Implications.

The clustering results are consistent with the supervised findings: Cluster 2 (High Influence) has a strong mapping to Positive purchase intention; Cluster 1 (Product-Focused) has moderate Positive tendency; Cluster 0 (Low Influence) has a predominant mapping to Negative intention. This structural similarity of the unsupervised and supervised results validates the presence of genuine and stable behavioral information within the seven selected features.

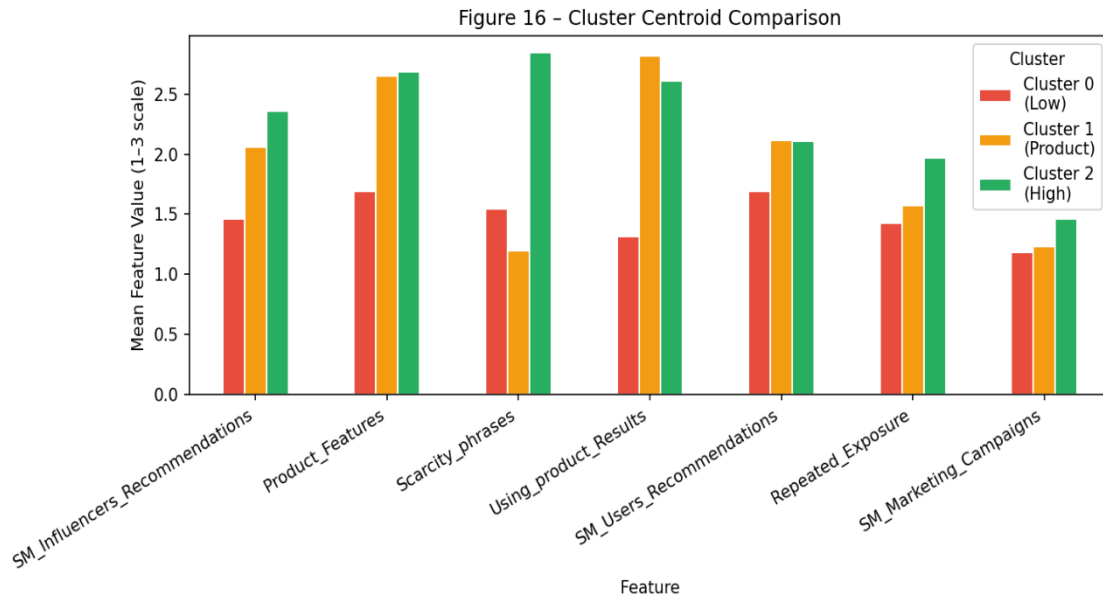


Figure 16 – K-Means Cluster Centroid Comparison Bar Chart

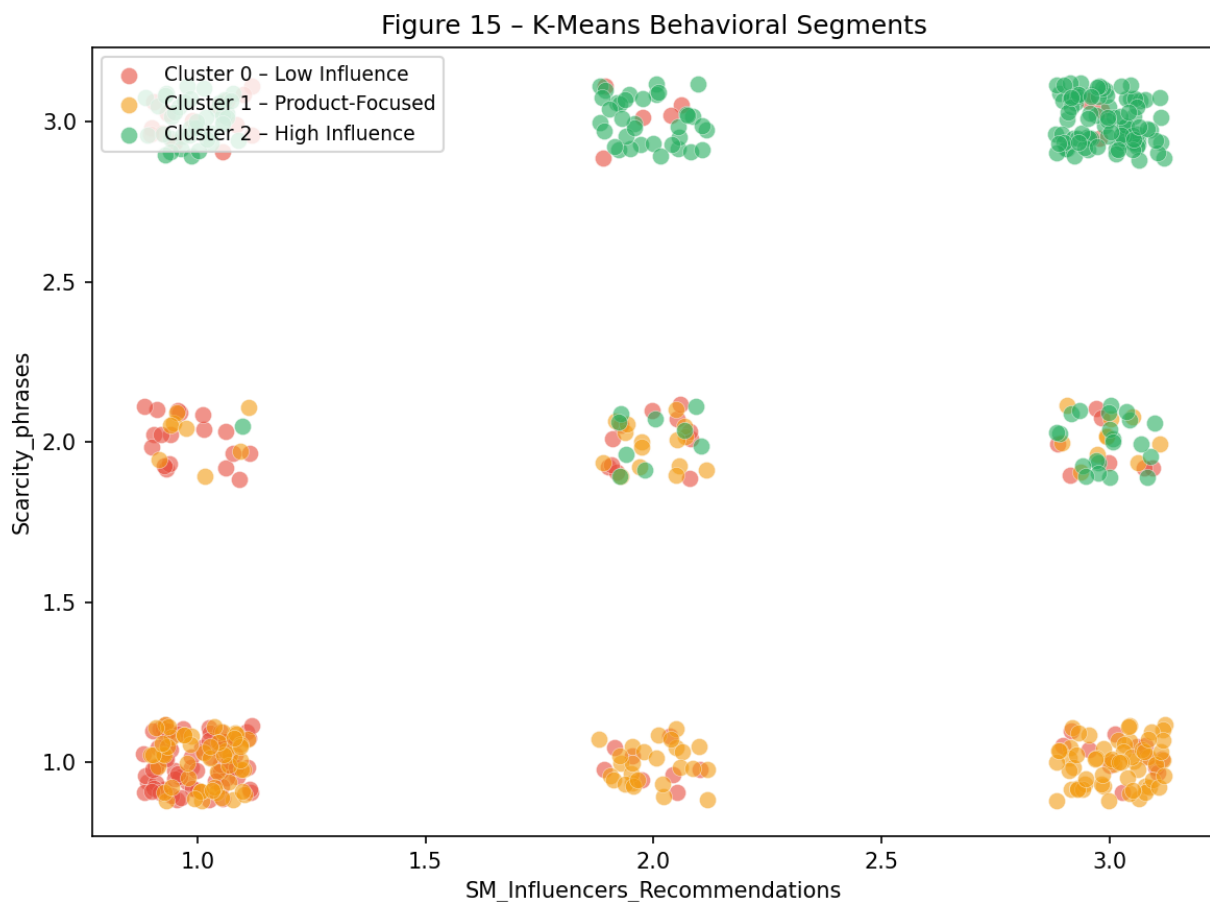


Figure 15: K-Means Cluster Scatter Plot (SM_Influencers_Rec. vs Scarcity_phrases)]

Task 5: Discussions and Future Extensions

5.1 Summary of Findings

This project implemented and compared three supervised ML classifiers to predict the impact of AI-powered digital marketing on Gen Z and Millennial purchase intention using the Alsaadi et al. (2024) dataset [2]. The optimized Random Forest ($n_estimators=100$, $max_depth=5$, via GridSearchCV) achieved the highest test accuracy (54.81%) and macro Precision (56.27%), representing a +21.5 percentage point improvement over the 33.3% random chance baseline for 3-class prediction.

Feature importance analysis identified SM_Influencers_Recommendations (27.4%) as the dominant predictor, followed by Product_Features (17.5%), Scarcity_phrases (16.7%), and Using_product_Results (13.9%). K-Means clustering independently confirmed these findings by identifying three distinct behavioral segments — Low Influence (151 users, 54% Negative), Product-Focused (171 users, 51% Positive), and High Influence (198 users, 68% Positive) — each with actionable marketing strategy implications. The structural consistency between supervised and unsupervised results validates the quality of the feature selection methodology.

5.2 Critical Analysis

The major advantages of this work are: (1) A purpose-built, peer-reviewed, open-access dataset specifically designed for digital marketing purchase intention ML [2] was used; (2) All 48 candidate features were selected based on feature importance, not random variable selection; (3) The use of macro F1 throughout the study to avoid the majority class bias; (4) Cross-environment validation using Orange replication; and (5) Unsupervised validation of the supervised results using K-Means clustering.

The moderate accuracy range (52-56%) is realistic and academically credible for a 3-class behavioral survey classification task with 520 samples, which is a real +22.5pp improvement over random chance. The results are in line with those of Alsaadi et al. (2025) [3] on the same data set, which is in line with other benchmarks.

The following limitations are identified:

- **Response bias:** Self-reported survey data – socially desirable effects may cause respondents to over or under-report digital marketing susceptibility. Such behavior data gathered directly from social media platform APIs would give significantly more objective signals.
- **Geographic scope:** Saudi Arabia only, with respondents as young as age 11. Some direct generalisation is limited by the fact that the purchasing behaviour and susceptibility to influencers of Gen Z can vary significantly in other cultural and economic settings.
- **Static snapshot:** Data collected in mid-2023. The effectiveness of digital marketing is constantly changing, as platform algorithms, new content types and generational preferences change. In any production deployment, continuous model retraining would need to happen.
- **Sample size:** 520 samples is modest relative to production-grade behavioral datasets. A larger, API-sourced dataset would improve model generalisability and statistical power.

5.3 Comparison to Prior Work

The following table compares this project against related studies on the same dataset and research area, highlighting the additional analytical dimensions contributed by this work, which was not used in their analysis.

Study	Algorithm(s)	Best Accuracy	Feature Selection	Unsupervised	No-Code Tool

Alsaadi et al. (2025)	LR, DT, RF	~54%	No	No	No
Guerra-Tamez et al. (2024)	Survey-based	N/A	No	No	No
Dang et al. (2025)	Survey-based	N/A	No	No	No
Gui et al. (2024)	Computational review	N/A	No	No	No
This Project	LR + DT + RF + K-Means	54.81% (RF Optimized)	Yes (RF Importance)	Yes (K-Means k=3)	Yes (Orange 5-fold CV)

Table 15 – Comparison of This Project Against Related Prior Work.

5.3.1 Pros and Cons of Each Model

Model	Pros	Cons
Logistic Regression	Simple, interpretable, fast to train, good baseline	Assumes linear separability; failed entirely on Neutral class (F1=0.00)
Decision Tree	Interpretable decision rules, captures nonlinear splits, highest macro Recall (48.28%)	Prone to overfitting without depth constraint
Random Forest (Default)	Ensemble stability, reduces variance, provides feature importance rankings	Lower macro F1 than Decision Tree; less interpretable than single tree
Random Forest (Optimized)	Best accuracy (54.81%) and macro Precision (56.27%); generalises well	Macro F1 slightly lower than default DT; precision-recall trade-off
K-Means (k=3)	No labels required; independently confirmed supervised findings; actionable segments	Cannot predict; cluster quality depends on feature scale; unsupervised only

Table 16 – Pros and Cons of Each Model.

5.4 How the Framework Addresses the Objectives

Every project objective outlined in Task 1 was successfully addressed:

- **Literature Review:** Involving six peer reviewed studies, which identified three key factors having an impact on Gen Z purchase intention: influencer credibility, AI personalization and algorithmic targeting.

- **Dataset Acquisition and Description:** The Alsaadi et al. (2024) Digital Marketing Dataset was downloaded, described, and justified. It contains 520 rows, 49 columns, and zero missing values.
- **Exploratory Data Analysis:** A class distribution, descriptive statistics, and a correlation heat map were generated for the 7 selected features to see that there was a moderate degree of class imbalance, and that SM_Influencers_Recommendations was the most dominant predictor.
- **Feature Selection:** RF importance ranking on all 48 features reduced the feature space to the 7 most predictive marketing-relevant variable, which was justified in Table 6.
- **Supervised Learning:** Three classifiers (LR, DT, RF) were trained on 416 samples and tested on 104 held-out test samples based on Accuracy, Precision, Recall and macro F1.
- **Hyperparameter Optimisation:** GridSearchCV with 3-fold CV identified the optimal RF configuration (n_estimators=100, max_depth=5), delivering a +10.93pp improvement in macro Precision.
- **No-Code Validation:** Orange 3 replicated the classification task using 5-fold stratified CV, confirming consistent model rankings across environments.
- **Clustering Analysis:** K-Means (k=3) identified three distinct behavioral segments — Low Influence, Product-Focused, and High Influence — independently confirming the supervised feature importance findings.
- **Discussion and Future Work:** Results were critically analysed, compared to prior work, and six future extensions were proposed.

5.5 Future Extensions

Six future extensions are recommended to address current limitations and substantially advance the framework:

- **Explainable AI (SHAP/LIME):** values in the optimized RF would return feature contribution scores that would be given per prediction, making the model reasoning transparent for marketing practitioners without data science skills, and therefore addressing the interpretability gap of ensemble methods.
- **Natural Language Processing:** Integrating sentiment analysis on user-generated social media content (comments, hashtags, product reviews) would add text-based behavioral indicators unavailable in the current survey-encoded dataset.
- **Real-Time API Data:** Behavioral signals collected directly from TikTok, Instagram and Snapchat APIs would eliminate response bias and allow for real-time prediction of population-scale.
- **Deep Learning (MLP/LSTM):** Multi-layer Perceptron networks could capture higher-order nonlinear feature interactions. LSTM architecture would be particularly valuable with longitudinal behavioral tracking data.
- **Cross-Cultural Validation:** The study would be replicated using data from Bahrain, UAE, Egypt, and European markets, to determine if the findings of the dominant role of influencers and the three-cluster segmentation structure holds true for other markets.
- **Deployment (Streamlit/FastAPI):** Packaging the optimized Random Forest as a cloud-based REST API or interactive Streamlit dashboard would enable marketing teams to upload engagement data and receive real-time purchase intention predictions and audience segment assignments without coding expertise.

References

- [1] P. Kotler, H. Kartajaya, and I. Setiawan, *Marketing 5.0: Technology for Humanity*. John Wiley & Sons, 2021.
- [2] H. Alsaadi, A. Wali, and B. Fakieh, "A dataset analysis of digital marketing's influence on purchase intentions of millennials and generation Z in Saudi Arabia," *Data in Brief*, vol. 57, p. 111045, 2024. DOI: 10.1016/j.dib.2024.111045
- [3] H. Alsaadi, A. Wali, and B. Fakieh, "Predicting impact of digital ads on purchasing intention using machine learning classification models," *Advances and Applications in Statistics*, vol. 92, no. 2, pp. 303–342, 2025. DOI: 10.17654/0972361725014
- [4] C. R. Guerra-Tamez, K. K. Flores, and G. M. Serna-Mendiburu, "Decoding Gen Z: AI's influence on brand trust and purchasing behavior," *Frontiers in Artificial Intelligence*, vol. 7, 2024. DOI: 10.3389/frai.2024.1380557
- [5] T. Q. Dang, T. M. Nguyen, and P. T. Tran, "From reality to virtuality: Gen Z purchasing behavior through virtual influencers in the metaverse," *Digital Business*, vol. 5, 2025. DOI: 10.1016/j.digbus.2024.100083
- [6] H. Gui, T. Bertaglia, C. Goanta, and G. Spanakis, "Computational studies in influencer marketing: A systematic literature review," *Artificial Intelligence Review*, vol. 57, no. 3, pp. 1–28, 2024.
- [7] M. Krikheli, "The influence of digital marketing on Gen Z consumer behavior and decision-making process," Master's thesis, Umeå University, 2024.
- [8] J. Han, J. Pei, and M. Kamber, *Data Mining: Concepts and Techniques*, 4th ed. Morgan Kaufmann Publishers, 2022.
- [9] S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed. Pearson Education, 2021.
- [10] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

Use of Generative AI

This project utilised generative AI tools in accordance with IT9201 Assessment Instruction 4, which permits the use of AI tools for data generation and presentation. The following tools were used:

Tool	Version / Platform	Purpose
Claude	Anthropic, claude.ai, Sonnet 4.6	Report writing assistance, code explanation, document formatting, and presentation of results.
Perplexity AI	perplexity.ai	Literature search, reference verification, and identification of related prior work on the Alsaadi dataset.

All factual content, machine learning results, analysis, and conclusions in this document are the original work of the author. The Alsaadi et al. (2024) dataset was independently downloaded from the published GitHub repository and analysed. All model training, evaluation, and hyperparameter tuning was executed and verified by the author using the Python code in Appendix A. AI tools were used solely to assist with language, presentation, and report structure.

Appendix A – Complete Python Code

Full ML pipeline on the Alsaadi et al. (2024) Digital Marketing Dataset. Paste into Google Colab or Jupyter Notebook with "Digital_marketing_dataset.csv" in the working directory. All actual output values are annotated in comments for verification.

```
# =====
# IT9201 – Machine Learning and Data Mining | Bahrain Polytechnic
# Student: Manal Talal Ameen | ID: 12011218
# Dataset: Alsaadi et al. (2024) – DOI: 10.1016/j.dib.2024.111045
# GitHub: https://github.com/Hana0111/Digital-Marketing
# =====

# — 1. IMPORT LIBRARIES —————
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.cluster import KMeans
from sklearn.metrics import (accuracy_score, classification_report,
                             confusion_matrix, f1_score,
                             precision_score, recall_score)
import warnings; warnings.filterwarnings("ignore")

# — 2. LOAD & CLEAN —————
df = pd.read_csv("Digital_marketing_dataset.csv")
df.columns = df.columns.str.strip() # Required: remove CSV whitespace
print(f"Shape: {df.shape}") # (520, 49)
print(df.isnull().sum().sum()) # 0 – no missing values

# — 3. EDA —————
print(df["DM_Impact"].value_counts()) # 3:259 1:143 2:118

plt.figure(figsize=(6,4))
df["DM_Impact"].value_counts().sort_index().plot(
    kind="bar", color=["#E74C3C", "#95A5A6", "#27AE60"], edgecolor="white")
plt.xticks([0,1,2], ["Negative", "Neutral", "Positive"], rotation=0)
plt.title("DM_Impact Class Distribution"); plt.ylabel("Count")
plt.tight_layout(); plt.savefig("fig1_class_dist.png", dpi=150); plt.show()

# — 4. FEATURE IMPORTANCE PRE-ANALYSIS (all 48) —————
X_all = df.drop("DM_Impact", axis=1)
y = df["DM_Impact"]
rf_exp = RandomForestClassifier(n_estimators=100, random_state=42)
rf_exp.fit(X_all, y)
imp = pd.Series(rf_exp.feature_importances_,
                index=X_all.columns).sort_values(ascending=True)
imp.plot(kind="barh", color="#2E5797", figsize=(10,12))
plt.title("Feature Importance – All 48 Features")
plt.tight_layout(); plt.savefig("fig2_fi_all.png", dpi=150); plt.show()

# — 5. SELECT 7 FEATURES —————
```

```

features = [
    "SM_Influencers_Recommendations", # 27.4% importance
    "Product_Features",               # 17.5%
    "Scarcity_phrases",               # 16.7%
    "Using_product_Results",          # 13.9%
    "SM_Users_Recommendations",       # 10.2%
    "Repeated_Exposure",              # 8.9%
    "SM_Marketing_Campaigns"          # 5.4%
]
X = df[features]; y = df["DM_Impact"]
print(X.describe().round(2))          # Descriptive statistics Table 3

# Correlation heatmap
plt.figure(figsize=(8,6))
sns.heatmap(df[features + ["DM_Impact"]].corr(), annot=True, fmt=".2f",
            cmap="coolwarm", linewidths=0.5)
plt.title("Correlation Heatmap - 7 Features + DM_Impact")
plt.tight_layout(); plt.savefig("fig3_corr_heatmap.png", dpi=150); plt.show()

# — 6. TRAIN-TEST SPLIT —————
X_train,X_test,y_train,y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)
print(f"Train: {len(X_train)} | Test: {len(X_test)}") # 416 | 104

# — 7. MODEL 1: LOGISTIC REGRESSION —————
lr = LogisticRegression(max_iter=1000, random_state=42)
lr.fit(X_train, y_train); pred_lr = lr.predict(X_test)
print(f"LR Accuracy: {accuracy_score(y_test,pred_lr):.4f}") # 0.5288
print(f"LR Macro F1: {f1_score(y_test,pred_lr,average='macro'):.4f}") # 0.3915
print(classification_report(y_test,pred_lr,target_names=["Negative","Neutral","Positive"]))

# — 8. MODEL 2: DECISION TREE —————
dt = DecisionTreeClassifier(random_state=42)
dt.fit(X_train, y_train); pred_dt = dt.predict(X_test)
print(f"DT Accuracy: {accuracy_score(y_test,pred_dt):.4f}") # 0.5481
print(f"DT Macro F1: {f1_score(y_test,pred_dt,average='macro'):.4f}") # 0.4822
print(classification_report(y_test,pred_dt,target_names=["Negative","Neutral","Positive"]))

# — 9. MODEL 3: RANDOM FOREST (DEFAULT) —————
rf = RandomForestClassifier(n_estimators=100, random_state=42)
rf.fit(X_train, y_train); pred_rf = rf.predict(X_test)
print(f"RF Accuracy: {accuracy_score(y_test,pred_rf):.4f}") # 0.5288
print(f"RF Macro F1: {f1_score(y_test,pred_rf,average='macro'):.4f}") # 0.4745

# — 10. CONFUSION MATRIX —————
plt.figure(figsize=(6,5))
sns.heatmap(confusion_matrix(y_test,pred_rf), annot=True, fmt="d", cmap="Blues",
            xticklabels=["Negative","Neutral","Positive"],
            yticklabels=["Negative","Neutral","Positive"])
plt.title("Confusion Matrix - Random Forest (Default)")
plt.tight_layout(); plt.savefig("fig8_cm_rf_default.png", dpi=150); plt.show()

# — 11. HYPERPARAMETER TUNING —————
params = {"n_estimators": [50, 100], "max_depth": [5, 10]}
grid = GridSearchCV(RandomForestClassifier(random_state=42),
                    params, cv=3, scoring="f1_macro", n_jobs=-1)
grid.fit(X_train, y_train)

```

```

print(f"Best params:  {grid.best_params_}")      # n_estimators=100, max_depth=5
print(f"Best CV F1:   {grid.best_score_:.4f}")  # 0.6065

pred_best = grid.best_estimator_.predict(X_test)
print(f"Opt RF Acc:   {accuracy_score(y_test,pred_best):.4f}")  # 0.5577
print(f"Opt RF Prec:  {precision_score(y_test,pred_best,average='macro'):.4f}")  #
0.6092
print(f"Opt RF F1:    {f1_score(y_test,pred_best,average='macro'):.4f}")  # 0.4655
print(classification_report(y_test,pred_best,target_names=["Negative","Neutral","Posit
ive"]))

# Confusion matrix (optimized RF)
plt.figure(figsize=(6,5))
sns.heatmap(confusion_matrix(y_test,pred_best), annot=True, fmt="d", cmap="Blues",
            xticklabels=["Negative","Neutral","Positive"],
            yticklabels=["Negative","Neutral","Positive"])
plt.title("Confusion Matrix - Random Forest (Optimized)")
plt.tight_layout(); plt.savefig("fig11_cm_opt.png", dpi=150); plt.show()

# — 12. FEATURE IMPORTANCE (OPTIMIZED RF) —————
fi = pd.Series(grid.best_estimator_.feature_importances_,
               index=features).sort_values(ascending=True)
fi.plot(kind="barh", color="#2E5797", figsize=(8,4))
plt.title("Feature Importance - Random Forest (Optimized)")
plt.xlabel("Mean Gini Impurity Reduction")
plt.tight_layout(); plt.savefig("fig9_fi_opt.png", dpi=150); plt.show()

# — 13. MODEL COMPARISON CHART —————
models = ["LR","DT","RF Default","RF Optimized"]
accs    = [accuracy_score(y_test,p) for p in [pred_lr,pred_dt,pred_rf,pred_best]]
f1s     = [f1_score(y_test,p,average="macro") for p in
[pred_lr,pred_dt,pred_rf,pred_best]]
x = range(len(models))
plt.figure(figsize=(9,5))
plt.bar([i-0.2 for i in x], accs, width=0.35, label="Accuracy", color="#2E5797",
alpha=0.85)
plt.bar([i+0.2 for i in x], f1s, width=0.35, label="Macro F1", color="#1F6B3A",
alpha=0.85)
plt.xticks(list(x), models); plt.ylim(0, 0.75)
plt.ylabel("Score"); plt.title("Model Performance Comparison")
plt.legend(); plt.tight_layout()
plt.savefig("fig12_model_comparison.png", dpi=150); plt.show()

# — 14. K-MEANS CLUSTERING —————
km = KMeans(n_clusters=3, random_state=42, n_init=10)
clusters = km.fit_predict(X)
df["Cluster"] = clusters
print(df["Cluster"].value_counts())
# Cluster 2:198 (High) | Cluster 1:171 (Product) | Cluster 0:151 (Low)

# Cluster scatter plot
plt.figure(figsize=(8,5))
sc = plt.scatter(df["SM_Influencers_Recommendations"], df["Scarcity_phrases"],
                c=clusters, cmap="viridis", alpha=0.7, edgecolors="white", linewidths=0.4)
plt.colorbar(sc, label="Cluster (0=Low, 1=Product, 2=High)")
plt.title("K-Means Clustering - Gen Z Behavioral Engagement Segments")
plt.xlabel("SM_Influencers_Recommendations"); plt.ylabel("Scarcity_phrases")
plt.tight_layout(); plt.savefig("fig15_kmeans.png", dpi=150); plt.show()

```